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Article

Conformational Chirality of Single-Crystal Covalent Organic Frameworks

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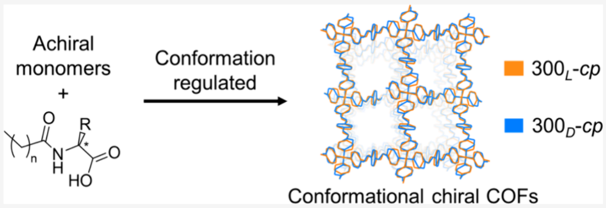
ABSTRACT: The crystallization of organic polymers is often hindered by chiral units, hence resulting in chiral organic polymers typically existing as amorphous or partially crystalline phases such as natural rubber and cellulose. Similarly, as an emerging crystalline chiral polymer, chiral covalent organic frameworks (COFs) also inevitably face a delicate balance between chiral units and crystallization, limiting their production and applications in separation, catalysis, and optics. Here, we present a general strategy for producing a series of conformational chiral COFs with high crystallinity through breaking the meso conformation of achiral COFs. Conformational chirality of COF-300 was constructed by involving chiral amino-acid derivative templates during synthesis and was proven to have excellent thermodynamic (200 °C annealing in air) and dynamic stability (61% cell volume change). The stereochemistry of the conformational chiral crystals can be controllably tuned by chiral templates, resulting in wide-range circular dichroism signals from ultraviolet to infrared wavelengths and absorption dissymmetry factors (g_{abs}) varying by up to 300%, with a maximum of g_{abs} = 0.012. This strategy paves the way for stereochemistry modification, property enhancement, and exploration of new applications of crystalline chiral materials.



■ INTRODUCTION

Chiral covalent organic frameworks (COFs) can be precisely designed and controlled by selecting appropriate chiral units,[1−8] making them a characteristic prototype for studying the structure−property relationships of chiral organic polymers in various applications including chiral separation, asymmetric catalysis, and chiral photonics.[7,9−11] However, the promise is limited by the crystallization of chiral COFs because their chiral units are rigid and asymmetrical (*R*/*S*). In contrast, flexible achiral molecules in the conventional sense are chiral in specific conformations in terms of stereochemistry (Figure 1a,b). Their conformations are here found to be able to self-adjust to meet the crystallization requirements by combining with chiral templates, resulting in highly crystalline conformational chiral COFs (Figure 1c). Moreover, their stereochemistry can be easily controlled by altering the chiral templates, thereby regulating their functionalities.

Herein, we selected the well-known COF-300,[12,13] which is achiral owing to its meso conformation (a molecule with multiple chiral conformations but lacking overall chirality due to the internal symmetry) as an example to showcase the construction of conformational chiral COFs. COF-300 exhibits a **dia** topology connected by benzaldehyde (BDA) and tetrakis(4-aminophenyl)methane (TAM) via imine bonds. The topological nodes of COF-300 are connected by linkers consisting of three benzene and two imine bonds (Figure 1b), which can be described with four dihedral angles φ_1, φ'_1, φ_2, and φ'_2, corresponding to four rotatable single bonds. Viewing along the single bond from imine to benzene, a *sinister* (*S*) configuration is noted when the in-between angle $\varphi \in (0°, 90°) \cup (-180°, -90°)$; otherwise, it is a *rectus* (*R*) configuration (Figure 1a). In previous studies,[12] it was revealed that COF-300 features an achiral structure in the space group $I4_1/a$, where the linkers display localized chiral conformation: *R* ($\varphi_1 = -23.3°$), *R* ($\varphi_2 = -2.9°$), *S* ($\varphi'_1 = 23.3°$) and *S* ($\varphi'_2 = 2.9°$), respectively (Supplementary Figure S1a,b). This is because flexible molecules prefer certain conformations for high-symmetry crystallization, forming meso conformations and counteracting the local chirality effects.

■ EXPERIMENTAL SECTION

Chiral templates guide the arrangement of molecules with flexible conformations with their asymmetric structure or surface, thereby controlling the stereochemistry of the products.[14−16] Here, chiral amino-acid derivatives were introduced as templates to break the structural meso conformation of achiral COF-300 to produce conformational chiral COFs. C_{16}-*L*-AlaA synthesized by condensing

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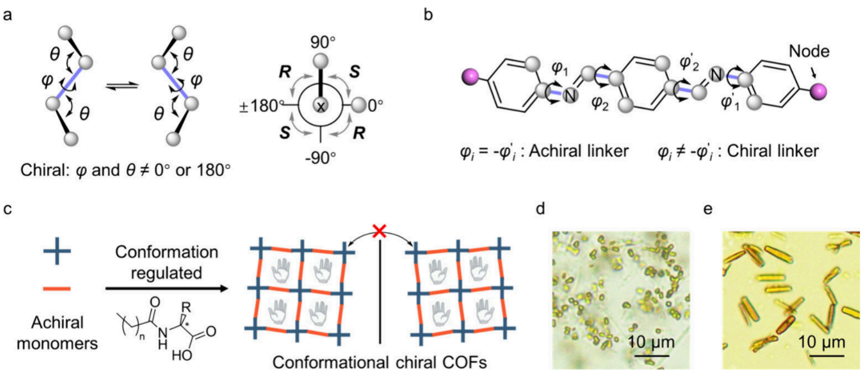

Figure 1. Conformational chirality of covalent organic framework. (a) Diagram depicting the chiral conformation of four connected atoms (left) and the relationship between the configuration of chiral conformations and dihedral angles (right). *R* and *S* refer to *rectus* and *sinister*, Latin for right and left, respectively. (b) Conformations of linkers in COF-300. $\varphi_i = -\varphi'_i$ corresponds to the achiral linker, while $\varphi_i \neq -\varphi'_i$ corresponds to the chiral linker. (c) Schematic diagram of conformational chiral COFs: amino-acid derivatives (R = $-CH_3$ or $-CH(CH_3)_2$, n = 10, 12, and 14) are used to regulate conformation. (d, e) Optical micrographs of conformational chiral COF-300 (d) and COF-304 (e).

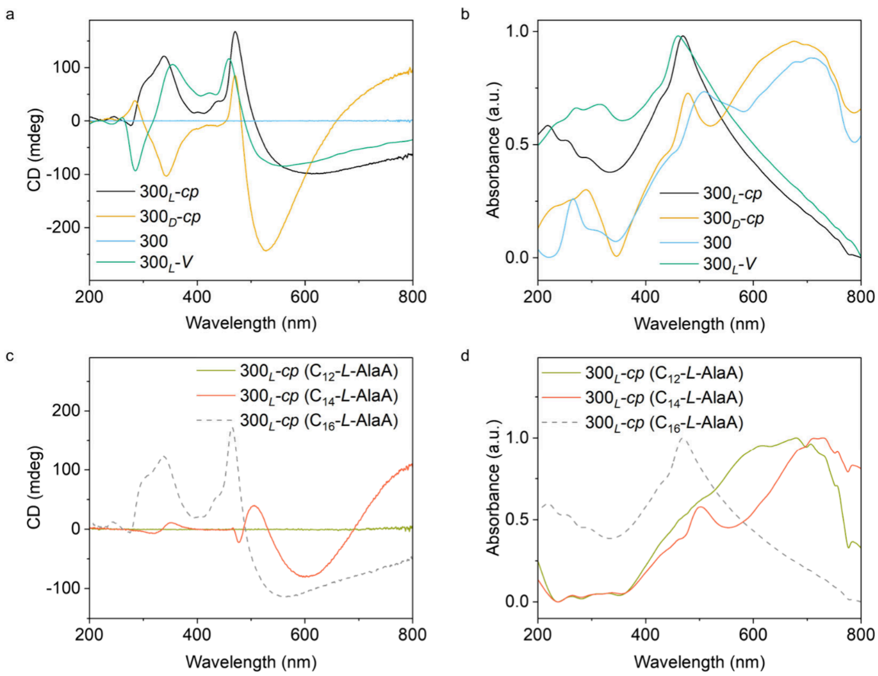

Figure 2. Chirality characterization. (a, b) Circular dichroism (CD) (a) and ultraviolet–visible (UV–vis) (b) absorption spectra of COF-300_L-*cp*, COF-300_D-*cp*, COF-300, and COF-300_L-*V*. (c, d) CD (c) and UV–vis (d) absorption spectra of COF-300_L-*cp* synthesized using amino-acid derivatives with different hydrophobic linker lengths (C_{12}-*L*-AlaA, C_{14}-*L*-AlaA and C_{16}-*L*-AlaA).

L-alanine with palmitoyl chloride[17] served as the chiral template (Figure 1c and Supplementary Section 1), forming chiral micelle dispersed in water (Supplementary Figure S2a). BDA and a solution of protonated TAM with *p*-toluenesulfonic acid were added to the reaction system subsequently, and crystals were formed within 1 h under ambient conditions (Figure 1d). Powder X-ray diffraction (PXRD) patterns were collected, and the diffraction peak at $2\theta = 6.4°$ corresponds to the (200) plane of opened COF-300, indicating the incorporation of chiral templates within the pores (Supplementary Figure S3a,b). This incorporation occurred during crystal growth, as chiral templates were adsorbed onto the crystal surface and then encapsulated within the pores. With three-dimensional electron diffraction (3D ED) analysis,[12,18] the unit cell parameters were determined to be a = 26.130(4) Å and c = 7.5900(15) Å, in the space group $I4_1$[19] (Supplementary Figure S1c) and all the non-hydrogen atoms were located (Supplementary Figure S3b). The resulting COF exhibits a **dia**-c7 topology (Supplementary Figure S3c), and it was designated as COF-300_L-*op* (*op* stands for opened phase).

RESULTS AND DISCUSSION

PXRD analysis showed that after washing with tetrahydrofuran and water, the 200 peak shifted to $2\theta = 9.0°$, indicating a contracted structure after the removal of chiral templates (Supplementary Figure S3a). All non-hydrogen atoms in COF-300_L-*cp* (*cp* means contracted phase) are solved by 3D ED, and then the accurate structure was obtained by Rietveld refinement against PXRD data with unit cell parameters a = 19.653(1) Å and c = 8.928(1) Å. The crystal structure revealed that channels were occupied by water molecules (Supplementary Figure S3d), confirming the removal of chiral templates. The circular dichroism (CD) spectrum of COF-300_L-*cp* exhibits positive signal peaks near 339 and 471 nm and a negative signal peak near 611 nm (Figure 2a), suggesting the presence of hierarchical chiral structures within the crystals.[20] The ultraviolet (UV)–visible absorption (Figure 2b) corre-

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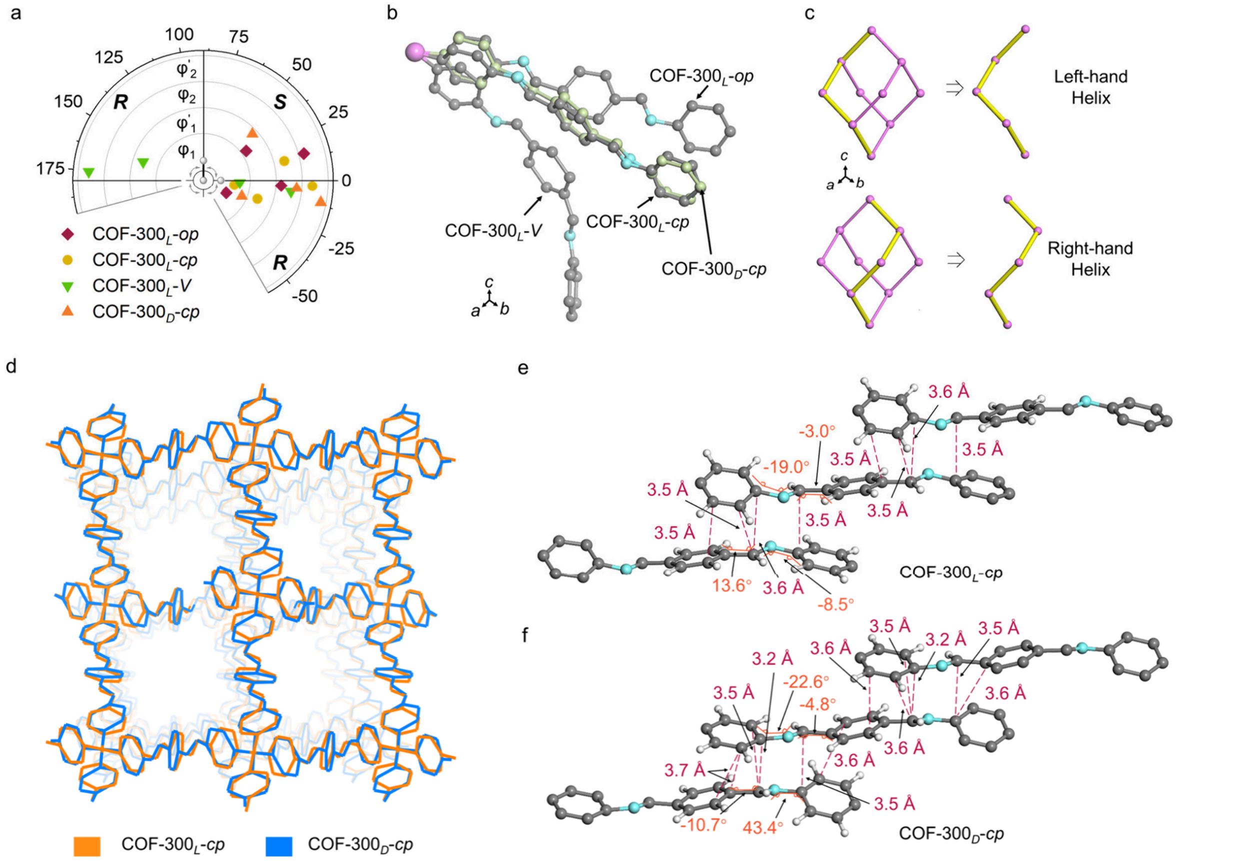

Figure 3. Crystal structure of chiral COFs. (a) Dihedral angles (φ_1, φ'_1, φ_2, φ'_2) of COF-300$_L$-*op*, COF-300$_L$-*cp*, COF-300$_L$-*V*, and COF-300$_D$-*cp*. (b) Stereochemistry of the linker varies across the crystal structures of COF-300$_L$-*op*, COF-300$_L$-*cp*, COF-300$_L$-*V*, and COF-300$_D$-*cp*, demonstrating the diverse conformations and spatial orientations of the linkers. (c) Schematic diagram of helical structures in diamond topology frameworks. (d) Crystal structure overlay of COF-300$_L$-*cp* and COF-300$_D$-*cp*. (e, f) Interframe interaction of COF-300$_L$-*cp* (e) and COF-300$_D$-*cp* (f) shows the origin of the conformation stabilizing.

sponding to n−π* and π−π* transitions of benzene and carbon−nitrogen double bonds indicates that the chirality of COF-300$_L$-*cp* comes from the framework composed of TAM and BDA. Conversely, achiral COF-300 exhibits no observable CD signal and distinct UV−visible absorption (Figure 2a,b) because of conformational variations (Supplementary Figure S3e).

The mechanism of chirality generation was clarified by monitoring the evolution of CD signal during the crystallization process and different heat treatments of chiral crystals, respectively. Before the characterization, all of the solid products were washed to remove the chiral templates (see Methods for details). Initially, the reaction of TAM and BDA yielded an amorphous phase without detectable CD signal within the first minute (Supplementary Figure S2b,c). As the reaction went on, the concentration of formed crystals increased, leading to a gradual intensity rise in the CD signal (Supplementary Figure S2c,d). These results indicate that the chirality transmitted by chiral templates cannot remain stable in the amorphous state but is maintained stably within crystalline structures. Furthermore, the thermal stability of conformational chirality was verified by various thermal treatments. The results show that the CD signals essentially remain during variable-temperature heating from 25 to 95 °C and thermal annealing treatments at 200 °C, highlighting the robust thermal stability of conformational chirality (Supplementary Figure S4a,b). However, the intensities of the signals gradually decrease with increasing temperature during hydrothermal treatment at 180 °C, concomitant with partial decomposition of the crystal (Supplementary Figure S4c,d).

To gain insight into the conformational chirality of the structure, we analyzed the geometrical conformation (Figure 3a,b) and helical structure (Figure 3c) of COF-300$_L$-*cp*. Within the chiral linker, the geometrical conformations are as follows: *R* (φ_1 = −8.5°), *R* (φ'_1 = −19.0°), *S* (φ_2 = 13.6°), and *R* (φ'_2 = −3.0°) (Figure 3a,b and Supplementary Table S1). Notably, due to the difficulty of absolute structure determination via 3D ED methods, the true structure of all chiral COFs in this work could be its enantiomer, but the missing the absolute structures will not influence the conclusion about the construction strategy of conformational chirality. The diamondoid framework consists of left- and right-handed helices along the *c*-axis, and each chiral linker is shared by adjacent helices (Figure 3c) with the same pitches of 62.5 Å (Supplementary Table S2). On the other hand, the distinct rotation of chiral linkers in opposite helix directions produces slightly different helix radii, 9.8 and 9.9 Å for the left- and right-handed helices, respectively (Supplementary Table S2).

Because of the interpenetration, the helices in COF-300$_L$-*cp* form one-dimensional pores along the *c*-axis (Supplementary Figure S3d), within which water molecules form helical water chains through hydrogen bonds. The water chains exhibit a left-handed configuration with a radius of 2.4 Å inside the right-handed helices and a right-handed configuration with a radius of 2.1 Å inside the left-handed helices (Supplementary Table S2). The different environments of the chiral pores made by left- and right-handed helices account for the variation in water chain radii. Generally, right-handed helices produce positive CD signals and *vice versa*.[20,21] However, the CD signals observed in chiral COF-300 originated from multiple factors, including interplay of the chiral conformations of the linker, helical structures, and other asymmetric factors, making it currently challenging to establish a one-to-one correspondence with its structure due to the complexity.

To understand the roles of different parameters on the stereochemistry of conformational chiral COFs, the influence of absolute configuration, hydrophobic chain length, head-group, and concentration of chiral templates were investigated. We first changed the absolute configuration of the chiral templates to obtain C_{16}-*D*-AlaA, which then served as the template for COF-300_D-*cp*. All non-hydrogen atoms in COF-300_D-*cp* are solved by 3D ED, and then the accurate structure was obtained by Rietveld refinement against PXRD data with unit cell parameters a = 19.682(1) Å and c = 8.933(1) Å, in the space group $I4_1$. The conformation of the chiral linkers was determined to be R ($\varphi_1 = -22.6°$), S ($\varphi'_1 = 43.4°$), R ($\varphi_2 = -4.8°$), and R ($\varphi'_2 = -10.7°$) (Figure 3a,b and Supplementary Table S1). As a result, comparing with COF-300_L-*cp*, the CD signals of COF-300_D-*cp* exhibit a reversal around 335 nm and a reduction at 471 and 527 nm (Figure 2a). Overlaying the frameworks of COF-300_D-*cp* and COF-300_L-*cp* reveals notable disparities in their local conformation (Figure 3d). The interframe distances ranging from 3.2 to 3.7 Å indicate strong $\pi-\pi$ interactions, which seems to play a crucial role in stabilizing the linkers' conformations (Figure 3e,f). As the interpenetrated framework expands, these interactions are significantly amplified, contributing to excellent thermodynamic stability of the conformational chiral structure (Supplementary Figure S4).

Next, we investigated the effects of the chiral templates with different hydrophobic chain lengths on the chirality of the COFs. The intensity, polarity inversion, and shift of the crystal's CD signals changed when the hydrophobic chain of the chiral template was shortened to 14 carbons (C_{14}-*L*-AlaA), and the CD signals were not detectable once the chain was further reduced to 12 carbons (C_{12}-*L*-AlaA) (Figure 2c,d). This is because the length of the hydrophobic chains in chiral templates affects their assembly in water where short hydrophobic chains are not conducive for the formation of assembly structures.[17] The effect of different amino-acid head groups is also assessed. Isopropyl groups were introduced to increase the steric hindrance of the chiral template (C_{16}-*L*-ValA), which caused the crystal's CD signal to decrease significantly and slightly red-shift (Supplementary Figure S5). The crystal's CD signal was almost eliminated by introducing a *tert*-butyl group (C_{16}-*L*-*t*BuA) to further increase the steric hindrance. This may be because a large steric hindrance at the amino-acid positions weakens the noncovalent interactions between the chiral template and the framework structure, hindering the transference of chirality.

We also changed the concentration of the C_{16}-*L*-AlaA micelle system during synthesis. The positive CD signal of the crystal at about 339 nm gradually weakened as the concentration decreased, while the strength of the crystal's CD signals at 465 and 557 nm gradually decreased and moved to a higher wavelength (Supplementary Figure S6a, b). There was a 3-fold decrease in the dissymmetry factor (g_{abs}), ranging from 0.012 to 0.004 (Supplementary Table S3). The UV−visible absorption peak shifted from 470 to 500 nm (Supplementary Figure S6c), indicating an increase in linker conjugation. This is because the size of the assembly formed from amino-acid derivatives decreases with a low concentration, corresponding to the gradual reduction of the cell volume (Supplementary Figure S6d). These findings demonstrate that the conformational chiral COFs exhibits controllable stereochemistry by tuning the chemical parameters of chiral template, which is beneficial for modification of their structure and optical activity. In contrast, modulating chiral structure through chemical structure alterations necessitates intricate and extensive screening of crystallization conditions, which is a relatively more complex procedure.[13,17,22]

The conformational variations induced by guest molecules were investigated to reveal the dynamic stereochemistry of the conformational chiral COFs. The unit cell volume of COF-300_L-*op* decreases by 50% after removing chiral templates, turning to hydrated COF-300_L-*cp* (Figure 4a). On the other

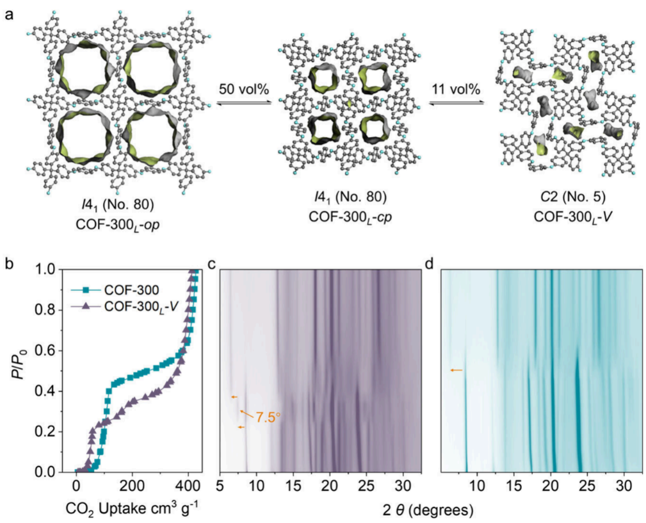

Figure 4. Dynamics of conformation chirality. (a) Crystal structures of the COF-300_L-*op* phase formed with chiral amino-acid derivatives, alongside the COF-300_L-*cp* phase postsubstitution of the chiral amino-acid derivative with water, and the COF-300_L-*V* phase after the removal of water. (b−d) CO_2 adsorption curves at 195 K (b) and the corresponding *in situ* powder X-ray diffraction (PXRD) patterns for COF-300_L-*V* (c) and achiral COF-300 (d).

hand, after the complete removal of water molecules, the space group of resulting COF-300_L-*V* changes from tetragonal $I4_1$ to monoclinic $C2$, with cell parameters of a = 26.21(1) Å, b = 8.089(4) Å, c = 19.69(2) Å, and β = 131.79(8)°, which is significantly different from results reported for activated achiral COF-300.[12,23] COF-300_L-*V* shows an additional 11% reduction in the unit cell volume in comparison with results for hydrated COF-300_L-*cp*, with linker conformations R ($\varphi_1 = -3.3°$), R ($\varphi'_1 = 162.6°$), R ($\varphi_2 = -7.4°$), and R ($\varphi'_2 = 175.5°$) (Figure 3a,b and Supplementary Table S1). Furthermore, the CD signals of COF-300_L-*V* shifted to 355 and 459 nm because of conformation variations, with inversion at 284 nm and attenuation at 527 nm (Figure 2a,b and Figure 3a). *In situ* PXRD data collected during CO_2 adsorption at 195 K (Figure 4b−d) show that COF-300_L-*V* was the initial form from vacuum to P/P_0 = 0.2, when its framework transformed into a mixture of COF-300_L-*V* and COF-300_L-*cp*. Between $P/$

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P_0 = 0.2 and 0.35, a new peak emerged around 2θ = 7.5°, corresponding to a previously unobserved intermediate state. A gradual transition from the mixing phase to COF-300$_L$-*cp* and then to COF-300$_L$-*op* can be detected at 0.35 < P/P_0 < 0.6 and 0.6 < P/P_0, successively. (Figure 4b,c). This guest-tagged structural transformation is different from that of achiral COF-300 (Figure 4b,d) and can be repeated without observable changes under repetitive gas/vapor absorption/desorption cycles (Supplementary Figure S7). Therefore, this phenomenon is significant evidence for maintaining dynamic stability of conformational chirality during the adsorption/desorption process, distinguished from flexible MOFs with multiple open structures, whose conformation changes according to various guest molecules.[24]

■ CONCLUSION AND PERSPECTIVE

Organic polymers typically face challenges in simultaneously possessing both chirality and high crystallinity. In this study, we develop highly crystalline, flexible conformational chiral COFs by stabilizing chiral conformations and inhibiting relaxation through precisely engineered lattice and reticular structures. Two conformational chiral COFs based on COF-300 and COF-304[25] (Figure 1e and Supplementary Figure S8) were successfully synthesized, verifying the general applicability of this strategy. Conformational chiral structures exhibit not only comparable stability, flexibility, and optical activity to other chiral materials consisting of rigid asymmetrical chiral units but also remarkably high crystallinity, which is typically found in noncovalent crystalline materials. The outstanding characteristics endow these materials with the potential for enhanced properties and a broad range of applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c13394.

> Synthesis procedures and analysis of chiral COFs; 3D electron diffraction (3D ED) analysis; powder XRD analysis; optical microscope images; thermal stability analysis; crystal size distribution analysis; infrared spectral analysis; solid-state nuclear magnetic resonance spectroscopic analysis (PDF)

Accession Codes

Deposition Numbers 2386536 and 2386596–2386598 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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